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Special Issue Article

An Affective–Cognitive Processing Model of Post-Traumatic Growth

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A topic that has begun to attract interest from clinical psychologists and psychotherapists is post-traumatic growth. First, we provide a general overview of the field, setting out the historical development, main concepts, measurement issues and research findings. Second, we review evidence showing that the relationship between post-traumatic stress and post-traumatic growth is likely curvilinear. Third, a new affective–cognitive processing model of post-traumatic growth will be introduced in which post-traumatic stress is understood to be the engine of post-traumatic growth. Fourth, points of clinical intervention are described showing the ways in which therapists can facilitate post-traumatic growth.
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Key Practitioner Message:
- No one approach to the treatment of trauma is able to help all people all of the time.
- Post-traumatic growth emerges out of satisfactory affective–cognitive processing.
- A common-factor approach that recognizes that there are many points of therapeutic intervention is described.

Keywords: Post-Traumatic Growth, Post-Traumatic Stress, Psychosocial Model, Organismic Valuing

A topic that has begun to attract interest from clinical psychologists and psychotherapists is post-traumatic growth—the study of how people often change in positive ways in their struggles with adversity. This paper provides a review of issues in post-traumatic growth research with the objective of developing new directions in theory, research and clinical practice. First, a general overview of the field will be provided, setting out the historical development, main concepts, measurement issues and research findings. Second, theory and evidence for the relationship between post-traumatic stress and post-traumatic growth will be reviewed showing that the relationship between the two is likely curvilinear. Third, a new affective–cognitive processing model will be introduced and the research and clinical implications considered.

To illustrate the concept of post-traumatic growth, consider the story of Anna, a patient of one of the authors (SR). Anna was a 26-year-old woman who was the victim of a violent rape by four men, whilst on vacation. Her boyfriend Stuart, who was with her at the time, was also badly physically assaulted in the attack. In the months and years that followed, Anna developed significant symptoms of post-traumatic stress. She had put on a significant amount of weight, began drinking heavily, had given up her job, developed a range of avoidance behaviour, could not stay alone in the house and had started sleeping with a carving knife under her pillow. By this time, she had married Stuart, who had been supportive throughout. Their feelings for each other had not changed as a result of their experience, if anything they had grown closer together. The subsequent birth of her daughter, Claire, prompted Anna to seek therapy. She spent 9 months in therapy and made significant progress bringing about a number of major changes in her life. There is, however, a tragic post-script to her story, which could be seen as illustrative of post-traumatic growth. Four years to the day following the assault, whilst on holiday with her family, Stuart dropped dead at the dining table at the cottage on the second day of their holiday. Although Anna quite naturally experienced a traumatic bereavement, over the years she has come through this, come to terms with Stuart's loss and has re-married. Although this brief vignette does not do full justice to her story or her resourcefulness in the face of overwhelming adversity, the following quotes in the succeeding

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paragraphs taken from a meeting some years following Stuart's loss can be seen as illustrative of post-traumatic growth.

> *. . .the rape was such a huge trauma. . . a huge thing—but I came through that and I thought after that I would never be able to live a normal life again ever. Then I was proved wrong because I got my life back to normality and then, with Stuart dying, it gave me belief in myself that I could come through that, and it has also made me think that life's very short and we have got to make the most of what you've got....I think over the last year, obviously there have been changes and my whole outlook on life is different because I had always thought I would have a big family, and I know that has changed now, but I can accept that that is the way it is now. I think I don't look as far into the future now, I just take.... Not every day as it comes, but just take short periods of time to see what happens, rather than planning the future. I think that has changed me as a person. . .Sometimes I think—I really can't believe that I am in this point in my life, laughing and happy—and even when I got out of the car today and walked up to see you—it is a long way from the time when I first came to see you and can't believe I come on my own and I wondered myself how I cope. I came because I have to, because I feel I have no choice, because I have got Claire who has to be able to live her life happily, and I also feel now—and I didn't always feel like this—that life is for living and I have got no choice. I am here, so I can either enjoy it and get on with it or be miserable.*

The story of Anna illustrates how trauma need not be a wholly destructive force in a person's life. In the end, the attack that Anna and Stuart experienced seemed to bring them closer together; and when Stuart died, Anna's previous experiences, rather than leaving her more vulnerable to the fact of his death, seemed to have given her a resilience—an ability to cope in the aftermath—that ultimately helped her become a person who has a deep appreciation of her life and is able to take responsibility over her choice how to live it.

Such stories will not be unusual to clinicians who work in the field of trauma. But until the advent of positive psychology, such stories of what we would now describe as post-traumatic growth were often overlooked as peripheral to the business of psychology. Previously, psychology had been largely, with the exception of the humanistic school (Maslow, 1943; Rogers, 1951), concerned with symptom reduction. As such, little was known about what led to a fulfilling and meaningful life. Fuelled by the emergence of positive psychology (Seligman & Csikszentmihalyi, 2000), the field of post-traumatic growth has now become established as a flagship topic for positive psychologists (Seligman, 2011) and a major field of enquiry attracting international attention from researchers, scholars and practitioners (Calhoun & Tedeschi, 2006; Joseph & Linley, 2008a; Weiss & Berger, 2010).

HISTORICAL DEVELOPMENT OF POST-TRAUMATIC GROWTH, ITS CONCEPTUALIZATION AND MEASUREMENT

Positive psychology has fuelled interest in post-traumatic growth over the past decade, but scientific interest in positive changes following adversity was sparked much earlier when a handful of studies appeared in the late 1980s and early 1990s, reporting positive changes in, e.g., rape survivors, male cardiac patients, bereaved adults, survivors of shipping disaster and combat veterans (see Joseph & Butler, 2010 for a review). Then, the topic of post-traumatic stress disorder (PTSD) was relatively new (following its introduction in 1980 by the American Psychiatric Association, 1980) and was attracting research interest. As such, the relatively few observations of positive change lay scattered throughout the literature overshadowed by research on the ways in which trauma could lead to the destruction and devastation of a person's life.

Interest took hold during the 1990s (see, e.g., O'Leary & Ickovics, 1995). Various terms were used throughout the literature to describe the positive changes that survivors' experienced, from, e.g., positive changes in outlook (Joseph, Williams, & Yule, 1993), stress-related growth (Park, Cohen, & Murch, 1996), thriving (Abraido-Lanza, Guier, & Colon, 1998) and perceived benefits (McMillen & Fisher, 1998). But most notably, it was the term *post-traumatic growth* (Tedeschi & Calhoun, 1996) that researchers gravitated to and has now become the term most widely used to describe this field of study and clinical practice.

Post-traumatic growth is a wide-ranging concept, still in development; but to date, three broad domains of positive change have been noted throughout the literature (Tedeschi & Calhoun, 1996). First, relationships are enhanced in some way. For example, people describe that they come to value their friends and family more and feel an increased sense of compassion for others and a longing for more intimate relationships. Second, people change their views of themselves in some way, e.g., that they have a greater sense of personal resiliency, wisdom and strength, perhaps coupled with a greater acceptance of their vulnerabilities and limitations. Third, people describe changes in their life philosophy, e.g., finding a fresh appreciation for each new day and re-evaluating their understanding of what really matters in life.

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Several self-report psychometric tools were developed during the 1990s to assess post-traumatic growth, notably the *Changes in Outlook Questionnaire* (Joseph et al., 1993), the *Perceived Benefit Scale* (PBS; McMillen & Fisher, 1998), *the Post-Traumatic Growth Inventory* (PTGI; Tedeschi & Calhoun, 1996); the *Stress-Related Growth Scale* (Park et al., 1996) and the *Thriving Scale* (TS; Abraido-Lanza et al., 1998). Although all measures assess aspects of the three broad dimensions noted above, they do so in a variety of ways with some offering unidimensional, and others multidimensional, solutions.

The PTGI has been the most widely used of the above aforementioned self-report measures. Several factor analytic investigations of its structure are now available. The original PTGI was developed on a sample of 604 college students. Principal component analysis revealed five factors that were labelled as relating to others, new possibilities, personal strength, spiritual change and appreciation of life. Several exploratory principal component analytic investigations of its structure have shown similar five-factor solutions (e.g., Maercker & Langner, 2001; Morris, Shakespeare-Finch, Rieck, & Newbury, 2005); but confirmatory factor analysis indicates that a single higher-order construct, with five first-order latent variables, provides an acceptable fit (Linley, Andrews, & Joseph, 2007).

The above results are informative about the psychometric properties of the PTGI. However, we must be cautious not to assume that these results apply more widely to understanding the structure of post-traumatic growth. To do this, a preferable strategy is to use several of the above measures simultaneously to assess a wider range of growthful experiences. In one such study, Joseph, Linley, and Harris (2005) conducted a principal component analysis of the three multifactorial measures, i.e., the five subscales of the PTGI, and the seven positive subscales of the PBS, and the eight subscales of the TS. Together, the PBS, PTGI and TS contain 20 positive change subscales. Three components had eigenvalues greater than 1.00. Inspection of the subscales loading on each provided evidence for three broad dimensions: (1) changes in perception of self; (2) changes in relationships with others; and (3) changes in philosophy of life. However, the eigenvalues-greater-than-one criterion is known to inflate the number of components to be extracted. Inspection of the scree plot showed only one component above a marked elbow; and thus, a forced one-component solution was also conducted, which showed that all subscales loaded highly on the component, providing further evidence that growth may be assessed as a unidimensional phenomenon.

But although much research has focused on the structure of extant psychometric tools such as the PTGI, the theoretical conceptualization of post-traumatic growth has received little attention. Unlike, e.g., PTSD, which has an agreed definition provided by DSM, which can provide the focus for measurement, there is no gold standard definition of post-traumatic growth. Each of the measures described above provides a particular and idiosyncratic operational definition. For example, although the PTGI is widely used, it is controversial because of its inclusion of religion as a defining feature (Joseph, 2011). Given the differences between the various measures, it is not surprising to find that they tend to be only moderately correlated and often differentially related to other personality and social psychology variables (see Joseph & Linley, 2008b for a review of measures). There is a need therefore for a gold standard conceptualization of post-traumatic growth, which can provide the basis for measurement development.

Clinical psychology has largely been concerned with subjective well-being, which can be broadly defined as emotional states. With positive psychology in the background, clinical psychologists are now also concerned with the facilitation of positive emotional states as well as their more traditional focus on the alleviation of negative emotional states. But post-traumatic growth does not refer to subjective well-being. Rather, as Joseph and Linley (2008b) suggest, post-traumatic growth refers to psychological well-being (PWB), which has traditionally been the focus of humanistic and existential psychologists. PWB is defined as high levels of autonomy, environmental mastery, positive relations with others, openness to personal growth, purpose in life and self-acceptance within contemporary literature (Ryff, 1989; Ryff & Singer, 1996).

The Psychological Well-Being Post-Traumatic Changes Questionnaire (PWB-PTCQ; Joseph et al., in press) was developed to assess post-traumatic growth as defined by an increase in PWB. The PWB-PTCQ provides a theoretically grounded operational definition that is more inclusive of aspects of change than previous measures.

Nonetheless, there is a limitation to the PWB-PTCQ in common to previous measures, which is that it relies on retrospective accounts of change—i.e., asking people to report on what positive changes they perceive themselves to have experienced since an event. As such, we might refer to this as perceptions of growth to distinguish those studies that assess actual growth as measured by calculating the difference between state measures of PWB before and after trauma.

Although there is no question that actual post-traumatic growth occurs, as this has been demonstrated in before-and-after studies (e.g., Peterson & Seligman, 2003), there is a question over how much actual growth correlates with perceptions of growth (Frazier et al., 2009). To investigate this issue, Frazier et al. (2009), in a longitudinal survey, compared actual changes with perceived changes finding only a weak correlation. Subsequent research suggests that the strength of association between actual and perceived

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change is moderated by the degree of distress, such that for those who are most distressed there is a weak correlation, but for those who are less distressed, there is a moderate correlation (Gunty et al., 2011). It may be that perceptions of growth are at times illusory and a way of coping with distress (Zoellner & Maercker, 2006). As such, researchers need to be wary of taking reports of growth at face value, particularly in the immediate aftermath of a crisis when people are most distressed.

Positive changes are widely reported by people following trauma. Using measures such as those described above, and open-ended interviews, a large number of studies have shown that growth is common for survivors of various traumatic events, including transportation accidents (shipping disasters, plane crashes, car accidents), natural disasters (hurricanes, earthquakes), interpersonal experiences (combat, rape, sexual assault, child abuse), medical problems (cancer, heart attack, brain injury, spinal cord injury, HIV/AIDS, leukaemia, rheumatoid arthritis, multiple sclerosis, illness) and other life experiences (relationship breakdown, parental divorce, bereavement, immigration). Typically, 30–70% of survivors will say that they have experienced positive changes of one form or another (Linley & Joseph, 2004).

Research indicated that greater post-traumatic growth is associated with personality factors such as emotional stability, extraversion, openness to experience, optimism and self-esteem; ways of coping such as acceptance, positive reframing, seeking social support, turning to religion and problem solving; and social support factors (Linley & Joseph, 2004). But as indicated above, there remain some pressing research issues over the measurement of growth and the use of instruments such as the PTGI, which ask about retrospective perceptions.

Sufficient cross-sectional research has now been conducted into the correlates of post-traumatic growth. What is now needed are prospective longitudinal studies able to document the development of growth over time, how both actual and perceived growth co-vary over time and how they relate to other variables—both as outcome variables to understand the development of growth and as predictor variables to understand the consequences of growth. More sophisticated theoretically informed designs are also called for in which we can begin to understand the interaction of personality, coping and social support variables in the prediction of growth and in its maintenance over time.

AFFECTIVE–COGNITIVE PROCESSING MODEL

Several theoretical approaches have been developed to account for post-traumatic growth (Joseph & Linley, 2006a), most notably the functional descriptive model of Calhoun and Tedeschi (2004), and the organismic valuing (OV) theory (Joseph & Linley, 2005). Both theories are based on Janoff-Bulman's (1992) theory of shattered assumptions—developed before the field of post-traumatic growth took hold—and propose that post-traumatic growth arises out of a person's cognitive struggle to resolve the challenged assumptive world. Both theories develop an understanding of how post-traumatic growth arises out of the interaction of personality, coping and social support variables, positing a proximal role for reflective rumination. These theories provide the basis for new research into the development of post-traumatic growth and its maintenance.

However, the most pressing theoretical issue is the relation between post-traumatic growth and post-traumatic stress. Consistent with the wider ambition of positive psychology to integrate negative and positive aspects of human experience (Joseph & Linley, 2006b), OV theory integrates a person-centred approach with Horowitz (1976) theory to suggest that post-traumatic growth is the normal and natural direction of successful working through. Horowitz describes an information processing approach in which working through involves alternating cycles of intrusion and avoidance.

Research prior to the development of the field of post-traumatic growth had already noted the ambiguous nature of intrusions—that although some studies showed greater frequency of intrusions to be related to poorer outcome, other studies showed greater frequency of intrusions to be related to better outcome (Creamer, Burgess, & Pattison, 1992). With the emergence of post-traumatic growth, this line of investigation has been explored further. Helgeson, Reynolds, and Tomich (2006) conducted a meta-analytic review concluding that greater post-traumatic growth was related to more intrusive and avoidant post-traumatic stress experiences.

As intrusion and avoidance are generally seen as symptoms of PTSD, at first glance, this result would seem to suggest that post-traumatic growth is indicative of poor mental health, but what Helgeson et al. suggest is that these constructs reflect cognitive processing:

> *Experiencing intrusive thoughts about a stressor may be a signal that people are working through the implications of the stressor for their lives, and these implications could lead to growth. In fact, some might argue that a period of contemplation and consideration of the stressor is necessary for growth to occur (p. 810).*

It is in this sense that post-traumatic stress—or at least the occurrence of intrusive and avoidant states thought to be indicative of working through—can be conceptualized as the engine of post-traumatic growth.

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However, it also seems that the relationship between post-traumatic growth and post-traumatic stress is a function of the intensity of post-traumatic stress. Butler et al. (2005), e.g., in their study following the attacks of September 2001, found that greater post-traumatic stress was associated with greater post-traumatic growth, but only up to a point, above which post-traumatic growth declines. It is suggested that there is a curvilinear relationship between post-traumatic stress and post-traumatic growth. Low levels of post-traumatic stress reactions indicate that the person has been minimally affected; thus, one would expect minimal post-traumatic growth. A moderate level of post-traumatic stress is indicative that the individual's assumptive world has in some way been challenged triggering the intrusive and avoidant experiences, but the person remains able to cope, think clearly and engage sufficiently in the necessary cognitive processing needed to work through. A high level of post-traumatic stress, however, where a diagnosis of PTSD might be considered is likely to mean that the person's coping ability is undermined, and their ability to cognitively process and work through their experience is impeded.

The inverted U-shaped relationship between post-traumatic stress and post-traumatic growth has been reported in several studies. For example, McCaslin et al. (2009) found a curvilinear relationship between post-traumatic stress and post-traumatic growth among 93 Sri Lankan university students who experienced a traumatic event; Colville and Cream (2009) reported a curvilinear relationship among 50 parents of children admitted to a paediatric intensive care unit; Kleim and Ehlers (2009) found a curvilinear relationship in two samples of 180 and 70 assault survivors, respectively; and Kunst (2010) found a curvilinear relationship among 678 victims of violence.

Within OV theory, the term post-traumatic growth is reserved exclusively for those positive changes that arise through the resolution of challenged assumptions as the person 'works through'. Defined in this specific way, post-traumatic growth should be distinguished from other benefits that people may report in the aftermath of adversity but which do not arise through challenged assumptions (Janoff-Bulman, 2004). For example, people may learn new coping skills through having to take on new roles. Davis and Nolen-Hoeksema (2009) use the term benefit finding in a distinct way to post-traumatic growth. Benefits can be found following a range of life events that are not necessarily traumatic; but in OV theory, the term post-traumatic growth is specifically used in the context of events that have led the person to experience some degree of post-traumatic stress (although not necessarily, but might have done, to such an extent that they would fulfil the diagnostic criteria for PTSD).

The greatest theoretical challenge to the field is to develop a new understanding of trauma that integrates post-traumatic stress and post-traumatic growth within a single conceptual framework that can guide clinical practice. It is as yet unclear which therapeutic models may most effectively facilitate post-traumatic growth. For example, a recent study indicated that cognitive–behavioural therapies that are known to alleviate PTSD did not have an effect on post-traumatic growth overall but nevertheless did have an impact on the domains of new possibilities and personal strength (Zoellner, Rable, Karl, & Maercker, 2011). It is clear that further research is needed into the elements of different therapeutic interventions that would most effectively facilitate post-traumatic growth. It is our view that not one therapy developed for the treatment of PTSD is likely sufficient to facilitate post-traumatic growth, which rather requires greater attention to therapeutic common factors and the multiple points at which help and advice can be offered.

To understand the different points at which help can be offered, Joseph, Williams, and Yule's (1997) psychosocial framework is based on three theoretical principles: (1) that subjective appraisal processes are the core of post-traumatic stress reactions and how meaning of an event is processed; (2) that the phenomena of post-traumatic stress are indicative of the need for cognitive processing and signify a natural and normal process; and (3) that personality, social and psychological factors influence the speed and depth of that processing (Joseph & Williams, 2005). As such, the psychosocial framework is consistent with the above research and theory on post-traumatic growth, and it is implicit in terms of the recognition that a rebuilding of the assumptive world takes place in light of appraisal processes (Joseph & Williams, 2005). Here, we modify the psychosocial framework to make the possibility of growth through revisions of the assumptive world now explicit in (Figure 1).

In this modified framework, affective–cognitive processing takes place via the cycle of event cognitions appraisal, emotional state, and coping as the person attempts to reconcile pre-trauma-related assumptions with the new trauma-related information. Consistent with OV theory, it is also proposed that people are intrinsically motivated to move towards growthful outcomes.

It is a non-medicalized framework that uses the language of social and cognitive psychology and views post-traumatic stress as a constellation of experiences that arise from event cognitions (memories), appraisal (intrusions), emotional state (negative affectivity) and coping (avoidance behaviours); but post-traumatic stress is viewed as normal and natural processes, *continuous* with normal behaviours, thoughts and emotions, rather than as a discontinuous state.

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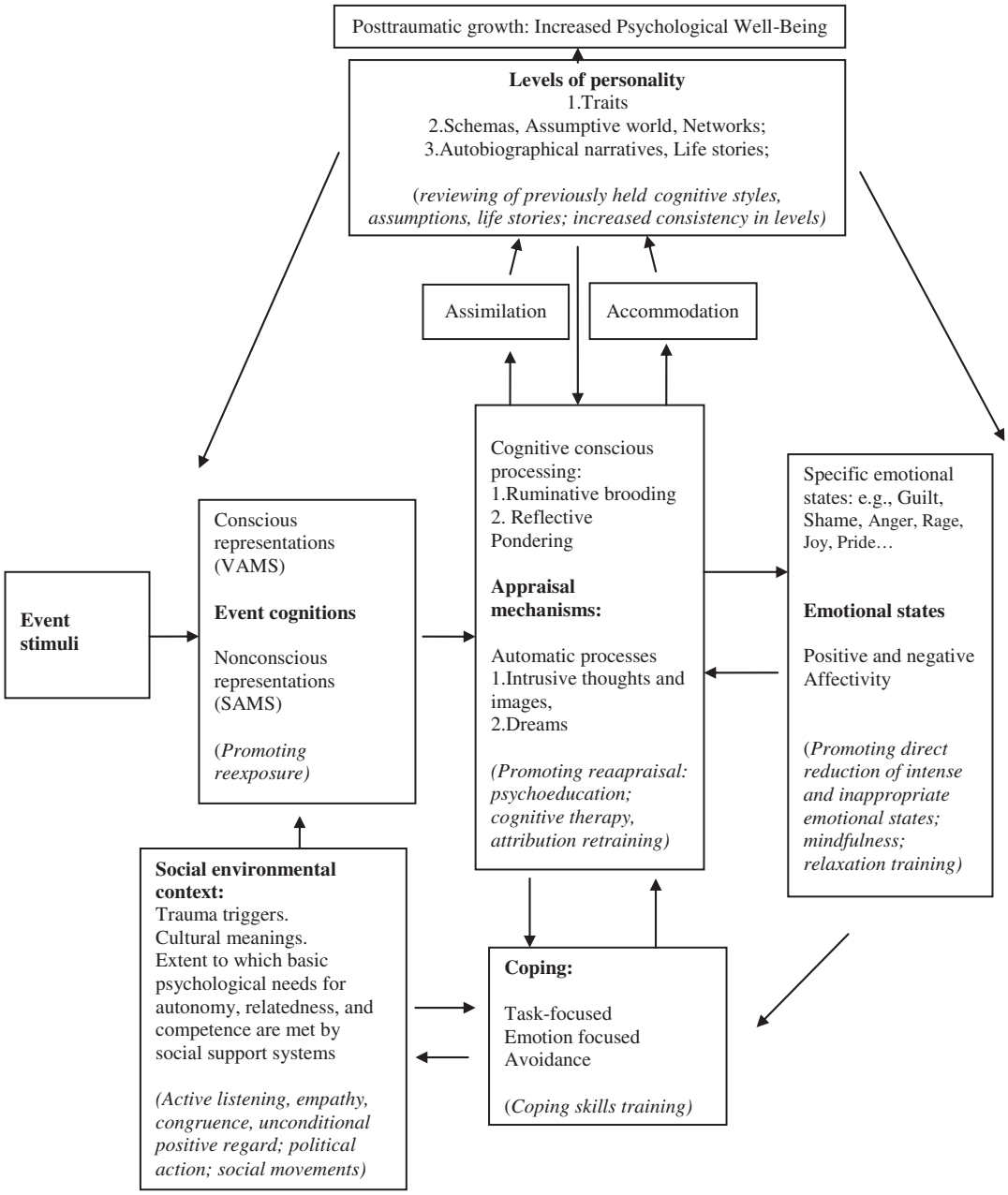

Figure 1. Post-traumatic affective–cognitive processing model

However, it is also recognized that, if the gestalt of the reformulated post-traumatic affective–cognitive processing model is not sufficiently available, the cycle of processing can become impeded giving rise to high and prolonged levels of post-traumatic stress. A person high on post-traumatic stress may be diagnosed with PTSD. However, the diagnosis does not necessarily indicate a dysfunction of mental mechanisms (as defined by Wakefield (1992) as when the mechanism is disordered in the sense of not functioning as it was 'designed to do by evolution'). In the following paragraphs, it is therefore assumed that brain mechanisms are able to perform their evolutionary derived functions.

To understand this framework, begin with the traumatic event and read clockwise through event cognitions, appraisals, emotional states and coping, as a repetitive

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cyclic process, which is ongoing until resolution of discrepancies between the pre-trauma assumptive world and the new trauma-related information are resolved. This process is influenced by personality and social environmental conditions (both broadly defined).

1. Event cognitions are subject to appraisal processes; the nature of which are determined by pre-existing assumptions about the self and the world, emotional states and coping activity. Events are experienced as traumatic to the person insofar as pre-existing assumptions are challenged by the new trauma-related information. For example, a senior police officer wrongly accused and charged for the alleged shoplifting of a trivial item experiences guilt and shame because of his or her subsequent treatment at the hands of colleagues. He or she feels humiliated, and this is a challenge to his or her status and position, previous exemplary professional record and sense of pride and honour. The realization that he or she may be thought capable of shoplifting is incompatible with his or her pre-event cognitive self-schema as honest and trustworthy. When there is an appraised incompatibility between the trauma-related information and pre-existing assumptions, affective–cognitive processing mechanisms are triggered.
2. Affective–cognitive processing is signified by the presence of intrusive thoughts, images and sensations, which become the subject of further conscious cognitive appraisal activity in the form of ruminative brooding (i.e., going over the information depicted but in a way that fails to find meaning and resolve discrepancies) and/or reflective pondering (i.e., thinking about the information depicted and its further meanings, drawing upon past experiences and prior assumptions). Ruminative brooding is often characterized by 'what if. . ..?' thoughts, which often play in a constant loop without a satisfactory resolution, thus often reinforcing negative emotional states such as guilt, shame or anger. Reflective pondering, however, might be expected to be growthful.
3. Cognitive appraisal activity leads to changes in negative emotional states, such as guilt, shame, anger, rage or positive emotional states such as hope, joy or pride, as well as states of general positive or negative affectivity.
4. Emotional states and general affectivity lead to a variety of coping strategies, so, e.g., guilt leads to the need to make reparation, shame to withdraw, anger to seek revenge, anxiety and fear fuels high levels of arousal and leads to cognitive and behavioural avoidance and so on. Numerous studies show that problem-focused and emotion-focused coping strategies are related to growth. Prati and Pietrantoni (2009) conducted a meta-analysis of 103 studies showing that optimism, social support, spirituality, acceptance coping, reappraisal coping, religious coping and seeking social support were associated with post-traumatic growth.
5. The feedback cycle of appraisal, emotional states and coping continues indefinitely driven by the need to affectively cognitive process the experience. The very process of coping becomes subject to further cognitive appraisal as the person learns about himself or herself and his or her ability to deal with adversity.
6. The speed and depth of processing is determined by social and environmental factors that determine the nature of the event cognitions and the coping strategies used.
7. Ruminative brooding prolongs the intrusion phase, but reflective pondering is necessary for the resolution of incompatibility between the new trauma-related information and pre-existing assumptive worlds. The shift from brooding to pondering is determined by the coping strategies and emotional states of the person. However, in some cases, high levels of physiological arousal, which in turn leads to a continued state of hypervigilance because of a perceived threat, even if the nature of the threat cannot be articulated, can impede this.
8. Reflective pondering allows for the person to begin to make meanings of his or her experience and to resolve discrepancies between the new trauma-related information and prior assumptive worlds (Stockton, Hunt, & Joseph, 2011). The importance of reflection is in narrative development (Meichenbaum, 2006). Tuval-Mashiach et al. (2004), e.g., examined narratives of five men who were exposed to a life-threatening terror attack showing that when their narratives were well built, coherent and possessing significance, levels of post-traumatic stress were lower. Reflection may be facilitated through a variety of means, including expressive writing, which is related to post-traumatic growth (Smyth, Hockemeyer, & Tulloch, 2008)
9. Discrepancies are resolved either through assimilative or accommodative processes. In assimilative processes, the person maintains or wishes to retain their pre-existing assumptions, often characterized by the self-narrative 'I want to be back to my old self'. This can be a defensive reaction in which the person has actively denied or distorted the new trauma-related information to achieve compatibility with existing assumptions. This has the potential for further traumatization in the wake of any further traumatic exposure as they have returned to their pre-existing trauma state and assumptions about the world have remained unchanged. In

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accommodation processes, the assumptive world is modified in light of the new trauma-related information, often characterized by the self-narrative 'I wouldn't be the person I am today if I hadn't gone through this'.

10. Accommodation involves a shift in the person's assumptive world to achieve congruence with the new trauma-related information and is understood within OV theory as the expression of an intrinsic drive towards PWB. Changes may be classified as being either negative (e.g., the world is a dangerous place where no one can be trusted) or positive (e.g., greater appreciation of family and friends, personal strength, the acceptance and validation of new meanings and outlooks on life and greater tolerance of others' shortcomings).

Negative accommodation and positive accommodation refer to the significance of the meanings attached to the accommodative process, and either can be growthful insofar as they are a corrective realistic balance to pre-existing assumptive worlds but can also be dysfunctional if they are 'over corrections'. This may in some cases encompass a struggle to find a natural balance, e.g., the person starts to recognise that the changes within himself or herself are not necessarily negative. This can be illustrated by the case of a man recovering from post-traumatic stress following a diving accident that involved significant subjective life threat. After struggling with symptoms and changes of meaning over 6 years, he decided to meet up with old university friends for a weekend 'get together'. After meeting up with some of the group, he eventually made an excuse and opted out of the weekend. He initially saw his actions as being a sign that he still had not regained his confidence in social settings. However, in an analysis of the experience later, he saw the reasons for his actions as having 'moved on' from the overly hedonistic and somewhat reckless activities and behaviours of that circle of friends and re-interpreted his actions in a more constructive and positive light.

CLINICAL PRACTICE

Turning to practice, the main implication of the above framework is that the task of the therapist is to provide a social environment that is able to support the continuous cycle of processing, on the understanding that when blocks are removed, affective–cognitive processing will follow. Clients may be distressed, but by itself, this is not indicative of impeded processing. Working collaboratively with the client over several sessions using the affective–cognitive model to understand the client's process will help identify if further therapeutic intervention may be helpful. Thus, there are several points at which intervention can be helpful (points of intervention are marked in Figure 1 in italics) involving:

1. Identifying with the client those social support processes that may be impeding processing: active empathic listening by the therapist may be sufficient to promote affective–cognitive processing; suggestions on obtaining other forms of support outside the therapy sessions may also be useful in helping promote affective–cognitive processing. Identification of sources of support that may impede processing may also be helpful.
2. Therapeutic work that engages the client in regular and consistent exposure-related activities to promote reappraisal of the traumatic experience and its meanings: by exposure we do not mean any one brand of therapy but any activities that allow for confrontation of the trauma-related information in a safe controlled environment. This includes talking and listening to structured exposure-based exercises. It may also involve actively engaging in the physical act of accompanying the client in a feared situation and supporting them through the experience, teaching them how to re-interpret environmental stimuli they have learnt to be fearful of.
3. Facilitating reappraisal: helping to normalize distressing emotional states, e.g., and to reappraise the meaning of the event and its significance within the person's life.
4. Promoting helpful coping strategies: e.g., supporting developing new coping skills, greater hopefulness, strategies for seeking social support and safe place imagery.
5. Reducing negative emotional states and promoting positive emotional states: e.g., relaxation exercises and gratitude exercises.

For people whose previous life experience has been unsupportive of their basic psychological needs and whose PWB was not already well developed prior to trauma, it may require attention to more points of intervention from the therapist to facilitate the process as the person lacks the developmentally acquired psychological mindedness.

The above clinical practice framework is able to integrate an understanding of post-traumatic stress and post-traumatic growth. Research is now needed to explore the utility of this approach. Particularly valuable will be qualitative research conducted over time, which is able to explore in depth the cycle of appraisals, emotional state factors, coping and their relation to assimilative and accommodative processes as the person struggles to rebuild his or her assumptions about the self and the world.

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CONCLUSION

Within our affective–cognitive processing theory, the term post-traumatic growth is used to describe the process of how positive changes arise in the aftermath of a traumatic event that challenges the person's assumptive world. There is much basic science yet needed to further our understanding of the architecture of post-traumatic growth, its predictors and means of facilitation. However, the research evidence has passed the point of questioning the validity of the construct of post-traumatic growth itself. There is no doubt that people do often change in terms of PWB following trauma. The greatest challenge for the field is to develop an integrative and clinically useful perspective. In this paper, we have presented some ideas for new research, directions for the future development of this field and an affective–cognitive processing framework to guide clinical practice.

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